159.102 Assignment 1 --- solution

There is always more than one possible solution. The following is one possible solution – it is not the only solution.

```
#include <iostream>
#include <cstdlib>  // need this for exit
using namespace std;

void binarytodecimal(string s);
void decimaltobinary(string s);
string convert(int num);

int main() {
  string str;
  cout << "Enter a number ";
  cin >> str;
  if (str[0] == '0') {
    binarytodecimal(str);
  } else {
    decimaltobinary(str);
  }
}

void binarytodecimal(string s) {
  int value, len, i;
  value = 0;
  len = s.length();
  if (len > 9) {
    cout << "This binary number has more than 9 binary digits.\n";
    exit(20);
  }
  for (i = 0; i < len; i++) {
    if ((s[i] == '0') || (s[i] == '1')) {
      value = value * 2 + s[i] - 48;
    } else {
      cout << "This is not a valid binary number.\n";
      exit(22);
    }
  }
  cout << "Converting binary to decimal. The result is ";
  cout << value << endl;
}

void decimaltobinary(string s) {
  string binary;
  int value, len, i;
  value = 0;
  len = s.length();
  for (i = 0; i < len; i++) {
    if ((s[i] >= '0') && (s[i] <= '9')) {
      value = value * 10 + s[i] - 48;
    } else {
      cout << "This is not a valid number.\n";
      exit(24);
    }
  }
  if ((value < 0) || (value > 255)) {
    cout << "This decimal number is outside the range 0 to 255.\n";
    exit(26);
```

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  }
  binary = convert(value);
  cout << "Converting decimal to binary.  The result is ";
  cout << binary << endl;
}

string convert(int num) {
// convert num to binary and display the result
  int len, i;
  char digit;
  string temp, result, display;
  i = 0;
  temp = "";
  while (num > 0) {
    digit = (num % 2) + 48;
    temp = temp + digit;
    num = num / 2;
  }
  // include extra zeros to get to 8 binary digits
  len = temp.length();
  while (len < 8) {
    temp = temp + "0";
    len = temp.length();
  }
  // reverse temp to get the result
  result = "";
  len = temp.length();
  for (i = len - 1; i >= 0; i--) {
    result = result + temp[i];
  }
  // place a space after first 4 digits
  display = "";
  for (i = 0; i < 4; i++) {
    display = display + result[i];
  }
  display = display + " ";
  for (i = 4; i < 8; i++) {
    display = display + result[i];
  }
  return display;
}
```